

Single Inheritance in Java

Code, Explanation, Test Cases and Output Screenshots

1. Program Code

```
package core_java;

class Animal {
    private String name;
    private int age;

    public Animal(String name, int age) {
        this.name = name;
        this.age = age;
    }

    public String getName() {
        return this.name;
    }

    public int getAge() {
        return this.age;
    }

    public void eat() {
        System.out.println(this.name + " is eating food.");
    }
}

class Dog extends Animal {
    private String food;

    public Dog(String name, int age, String food) {
        super(name, age);
        this.food = food;
    }

    public void showDetails() {
        System.out.println(getName() + " is " + getAge()
            + " years old and eats " + this.food);
    }
}

public class SingleInheritance {
    public static void main(String[] args) {
        Dog D = new Dog("Buddy", 3, "Chicken");

        D.eat();
        D.showDetails();
    }
}
```

2. Explanation

Animal Class: Animal is the parent class. It stores the name and age of an animal. The constructor initializes these values, while getName() and getAge() return them. The eat() method displays an eating message.

Dog Class: Dog is the child class that extends Animal. This demonstrates single inheritance because Dog inherits features from one parent class.

super Keyword: The Dog constructor uses super(name, age) to initialize the name and age in the Animal class.

showDetails() Method: This method uses the inherited getter methods and displays the dog's name, age, and food.

main() Method: A Dog object is created. The inherited eat() method and the showDetails() method are then executed.

3. Test Cases

Test Case	Values Used	Expected Output	Verification
Test Case 1	Dog("Buddy", 3, "Chicken")	Buddy is eating food. Buddy is 3 years old and eats Chicken	PASS
Test Case 2	Dog("Max", 5, "Rice")	Max is eating food. Max is 5 years old and eats Rice	PASS
Test Case 3	Dog("Rocky", 2, "Meat")	Rocky is eating food. Rocky is 2 years old and eats Meat	PASS

4. Output Screenshots

The screenshots below contain only the actual output. Test case labels and input values are kept outside the screenshots.

Test Case 1 – Values: Dog("Buddy", 3, "Chicken")

```
Buddy is eating food.
```

```
Buddy is 3 years old and eats Chicken
```

Test Case 2 – Values: Dog("Max", 5, "Rice")

```
Max is eating food.
```

```
Max is 5 years old and eats Rice
```

Test Case 3 – Values: Dog("Rocky", 2, "Meat")

```
Rocky is eating food.
```

```
Rocky is 2 years old and eats Meat
```

5. Result Verification

All three test cases produce the expected output. The Dog class successfully inherits behavior from the Animal class, confirming that single inheritance is working correctly.